IN SILICO PREDICTION OF DEGRADATION PATHWAY AND COMPARISON WITH REAL TIME DATA

Project report Submitted to St Joseph's University, Bengaluru

In partial fulfillment of requirement for the Award of Master's Degree in Organic Chemistry

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DECLARATION

I, **YANA A B**, hereby declare that the M.Sc. dissertation entitled **"In Silico Prediction of Degradation Pathway and Comparison with Real Time Data"** incorporates the result of my independent research under the supervision of **Dr. Manoj Kumar Singh,** Microlabs Kudlu, and **Dr. Anchal Singhal**, Department of Chemistry, St Joseph's University, Bengaluru.

I further declare that it has not been previously submitted in full to this or to any other university for any degree or diploma. Due acknowledgements have been made whenever content has been cited from other sources.

Date: May, 2026

Place: Bengaluru

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CERTIFICATE

This is to certify that the M.Sc dissertation entitled, "**In Silico Prediction of Degradation Pathway and Comparison with Real Time Data**" submitted by **YANA A B (241CHO15)** in fulfillment of the requirements for the award of Master's Degree in Organic Chemistry, is an original work. I certify that this has not been submitted in full to any university for the award of any degree or diploma.

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ABSTRACT

Pharmaceutical stability studies are essential for ensuring the safety, efficacy, and quality of drug products; however, conventional methods are time-consuming and resource-intensive. This study focuses on in silico prediction of degradation pathways and their comparison with experimental data to enable faster, more efficient stability assessment.

Initial degradation data obtained over **6 months** at different temperatures were used to model and **extrapolate impurity levels up to 22 months** using zero-order kinetics. Both **Excel-based calculations and AI-based linear regression models** were applied, and the predicted results showed strong agreement with experimental trends, validating the extrapolation approach.

The **Arrhenius equation** was used to calculate the activation energy (E_a) under different pH conditions. The similarity in activation energy values indicated a **consistent degradation mechanism**, unaffected significantly by pH variations.

Additionally, **Zeneth** was employed to predict degradation pathways under varying temperature and pH conditions. The results showed a **single dominant pathway with high pathway scores**, which closely matched the experimental findings.

Overall, this study demonstrates that integrating **kinetic modeling, AI-based prediction, and in silico tools** can reliably predict pharmaceutical degradation behavior. The strong correlation between predicted and experimental data confirms that such approaches can significantly reduce the time required for stability studies while ensuring accurate pathway identification and validation.

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1. Introduction:

Pharmaceutical stability studies are essential to ensure drug safety, quality, and effectiveness throughout their shelf life. Evaluating degradation behavior under different conditions helps **predict impurity formation, determine storage requirements, and meet regulatory standards**. This work focuses on applying **kinetic analysis, stability testing principles, and predictive tools to understand and model drug degradation.**

1.1 Background of Pharmaceutical Stability:

Pharmaceutical stability refers to the ability of a drug substance or drug product to maintain its physical, chemical, microbiological, therapeutic, and toxicological properties within specified limits throughout its shelf life. Stability is a critical quality attribute that ensures the safety and efficacy of pharmaceutical products from manufacture through consumption. To ensure product quality and stability, studies are conducted under defined conditions in accordance with the **International Council for Harmonization (ICH) guidelines, particularly ICH Q1A(R2).** These studies help determine shelf life, recommended storage conditions, and packaging requirements. Traditionally, stability testing is a time-consuming process, often requiring months to years to generate sufficient data**. Accelerated stability studies are used to predict long-term behavior by exposing drugs to elevated stress conditions**. However, these approaches may not always accurately capture complex degradation pathways.

1.2 Factors & Mechanisms of Degradation:

Pharmaceutical degradation refers to the chemical or physical changes that occur in a drug substance or product over time, leading to reduced efficacy and possible formation of harmful impurities. Understanding both the factors that influence degradation and the underlying mechanisms is essential to ensuring drug stability and safety.

Factors Affecting Degradation

Drug degradation is influenced by several environmental and formulation-related factors:

- **Temperature:** Elevated temperatures accelerate degradation reactions by increasing molecular kinetic energy, often following Arrhenius behavior.
- **Humidity (Moisture):** Water can promote hydrolytic reactions, especially in drugs containing ester, amide, or lactam groups.

- **Light Exposure:** Ultraviolet (UV) and visible light can induce photochemical reactions, leading to photodegradation.
- **Oxygen:** Oxidative degradation occurs in the presence of oxygen, particularly in drugs susceptible to free radical reactions.
- **pH:** The stability of many drugs is highly dependent on pH, influencing the rate of hydrolysis and other reactions.
- **Excipients & Formulation:** Interactions with excipients or incompatibilities can accelerate degradation.
- **Packaging & Storage Conditions:** Improper packaging may expose drugs to environmental stress, affecting stability.

Mechanisms of Degradation

Pharmaceutical degradation typically occurs through well-defined chemical pathways:

- **Hydrolysis:** One of the most common mechanisms, involving the reaction of a drug with water, leading to bond cleavage (e.g., esters and amides).
- **Oxidation:** Involves loss of electrons or reaction with oxygen, often forming peroxides or free radicals. Drugs with phenolic or unsaturated structures are especially prone.
- **Photolysis:** Degradation caused by light exposure, resulting in structural changes or bond breakage.
- **Isomerization:** Conversion of a drug into a different isomer, which may have reduced or no therapeutic activity.
- **Thermal Degradation:** Decomposition due to heat, which may accelerate other degradation pathways.
- **Reduction:** Gain of electrons leading to structural changes, depending on the chemical environment.

Understanding these factors and mechanisms is crucial for designing stability studies, selecting appropriate formulations, and **predicting shelf life**. It also serves as a foundation for developing predictive models, such as **AI-based tools, to estimate degradation pathways and impurity formation.**

1.3 Forced Degradation Studies:

Forced degradation studies involve exposing a drug to extreme conditions, such as heat, light, humidity, acidic/basic environments, and oxidative conditions, to accelerate its breakdown. These studies help identify **degradation pathways, impurity formation,** and ensure **the development of stable and safe pharmaceutical products,** as recommended by the International Council for Harmonization guidelines.

1.4 Kinetics & Activation Energy:

Pharmaceutical degradation kinetics describes how the rate of drug degradation changes over time and under different conditions, typically following zero-order **or** first-order kinetics**.** Studying these kinetics allows estimation of rate constants (k), half-life ($t_{½}$), and shelf life (t90), which are critical for stability prediction.

The temperature dependence of degradation is explained using the Arrhenius equation, which relates the rate constant (k) to temperature. **By plotting ln k versus 1/T, a straight line is obtained, and the slope is used to calculate** activation energy (Ea)**.** This helps in extrapolating stability.

Activation energy represents the minimum energy required for degradation to occur. **A higher Ea indicates slower degradation and better stability.** This concept is widely used in predictive modeling and was key to this study, enabling extrapolation of degradation behavior **and estimation of long-term stability from short-term experimental data.**

1.5 In Silico Prediction & Tools:

In silico prediction tools are computational methods used to **estimate drug degradation pathways, impurity formation, and stability without relying entirely on long-term experimental studies.** These approaches help reduce time, cost, and effort in pharmaceutical development.

One key tool is Zeneth, which predicts chemical degradation pathways based on known reaction mechanisms and the molecule's structural information. **Zeneth analyzes the drug's structure and simulates its degradation under various conditions, including hydrolytic, oxidative, and thermal stress.**

It provides insights into likely degradation routes, identifies potential impurities, and helps confirm

whether a consistent degradation pathway is followed. In this study, **Zeneth was used to support and validate the experimentally observed degradation behavior, enhancing confidence in the predictive model.**

Overall, in silico tools like Zeneth play an important role in modern pharmaceutical research by enabling faster and more efficient stability prediction.

2. Literature survey

Pharmaceutical stability studies are essential to ensure the safety, efficacy, and quality of drug products throughout their shelf life. These studies are systematically guided by regulatory frameworks that define testing conditions, data interpretation, and storage requirements. A strong foundation in stability guidelines and degradation kinetics is crucial for developing predictive models and understanding long-term drug behavior.

ICH Guidelines related to Stability Studies:

- **ICH Q1A:** General guidelines for stability testing of new drug substances and products
- **ICH Q1B:** Photostability testing to evaluate the effect of light exposure
- **ICH Q1C:** Stability testing requirements for new dosage forms
- **ICH Q1D:** Bracketing and matrixing designs to reduce the number of stability studies
- **ICH Q1E:** Evaluation and statistical interpretation of stability data
- **ICH Q1F:** Stability data requirements for different climatic zones (I: Temperate, II: Mediterranean/Subtropical, III: Hot/Dry, IVa/IVb: Hot/Humid)

Several research studies have contributed to understanding degradation kinetics and the application of Arrhenius principles in pharmaceutical stability. These works provided a theoretical and practical basis for integrating predictive modeling into this project.

Key Learnings from Literature:

- **P. A. MacFaul et al. (2011):**
    - Activation energy (Ea) is a critical parameter for predicting hydrolytic stability
    - Arrhenius-based analysis helps identify degradation trends across drug-like molecules

- Useful for early-stage prediction of stability without extensive long-term data

- **Mohamed H. Abdel-Hay et al. (2019):**
  - Demonstrated calculation of degradation rate constants using Arrhenius plots
  - Showed how forced degradation data can be used to estimate activation energy
  - Highlighted differences in stability between compounds under similar conditions
- **K. Tamura et al. (2020):**
  - Introduced modified Arrhenius equation incorporating excipient effects
  - Showed improved prediction accuracy in real formulation conditions
  - Emphasized the importance of formulation factors in degradation kinetics
- **S. J. Rajput et al. (2021):**
  - Evaluated degradation kinetics across different pH and temperature conditions
  - Demonstrated calculation of kinetic parameters like Ea and rate constants
  - Highlighted the combined impact of environmental factors on drug stability

These studies collectively supported the application of Arrhenius kinetics and stability modeling in this project, enabling accurate prediction of degradation behavior and long-term stability from accelerated data.

3. Methodology

3.1 Data Collection:

The primary data required for this study were obtained from the pharmaceutical company and included **the initial and final concentrations of the drug to assess the extent of degradation.** In addition, a detailed dataset was collected that shows the progressive degradation profile over the first six months. This dataset included information on drug concentration at different time intervals (monthly) under defined temperature conditions. **Such structured data were essential for understanding the degradation pattern and for applying kinetic models.** The availability of early-stage degradation data also supported extrapolation to longer time periods **using Arrhenius-based approaches, forming the basis for predictive stability analysis.**

N1003		
	% of Impurity	
Time Point	Temp. 40°C	Temp. 25°C
0	0.05	0.05
3	0.7	0.2
6	1.4	0.3

3.2 Kinetic Analysis & Extrapolation:

Kinetic analysis was performed to understand the rate and pattern of drug degradation and to enable the prediction of long-term stability using mathematical models. Both first-order and zero-order kinetics were evaluated based on the nature of the degradation data.

First-Order Kinetics

$k = 2.303/t * \log(C0/Ct)$

- **k** = rate constant
- **t** = time
- $\mathbf{C_0}$ = initial concentration of drug
- $\mathbf{C_t}$ = concentration at time *t*

This equation is derived from the integrated first-order rate law, in which the rate of degradation depends on the drug concentration. It was used to calculate the degradation rate constant from experimental data.

Arrhenius Equation (Temperature Dependence)

$\ln k = \ln A - Ea / RT$

- **k** = rate constant
- **A** = frequency factor (collision frequency)
- $\mathbf{E_a}$ = activation energy
- **R** = gas constant
- **T** = temperature (in Kelvin)

This equation explains how temperature affects the rate of degradation. By plotting **ln k vs 1/T**, a straight line is obtained:

From the slope, activation energy (E_a) was calculated. This is crucial **because higher $\mathbf{E_a}$ indicates greater stability and allows extrapolation of degradation rates at lower temperatures from accelerated conditions.**

$$\text{slope} = -Ea/R$$

Zero-Order Kinetics (Linear Degradation Model)

Ct = m*months

- $\mathbf{C_t}$ = concentration (or impurity level) at time *t*
- **m** = slope (rate of impurity increase)
- **t** = time (months)

This model is based on the zero-order rate law, in which degradation occurs at a constant rate independent of concentration. A linear increase in impurity vs time confirms zero-order behavior.

Determination of Reaction Order

- If a plot of concentration vs time is linear → Zero-order
- If a plot of log concentration vs time is linear → First-order

The best-fit model (highest linearity) was selected to describe the drug's degradation behavior.

Zero-Order Rate Constant

k = {C_0 - C_t}/{t}

- **k** = rate constant
- $\mathbf{C_0 - C_t}$ = change in concentration
- **t** = time

If the impurity increases over time, the rate is considered positive, indicating degradation progression.

Extrapolation & Shelf-Life Prediction

Each drug has a **specified impurity limit** that must not be exceeded. Using the degradation model, extrapolation helps predict the **time (month)** at which this limit will be crossed.

- By extending the degradation curve beyond observed data
- The intersection with the **acceptable impurity limit** gives the predicted shelf life

This allows estimation of long-term stability using short-term experimental data.

Activation Energy-Based Rate Prediction

ln{k_{60}}/{k{25}} = {E_a}/{R}({1}/{T{25}} - {1}/{T{60}}

- $\mathbf{k_{60}}$, $\mathbf{k_{25}}$ = rate constants at 60°C and 25°C
- $\mathbf{E_a}$ = activation energy
- **R** = gas constant
- **T** = temperature in Kelvin

This rearranged Arrhenius equation was used to calculate **unknown rate constants at lower temperatures** using high-temperature data. It serves as the basis for **accelerated stability studies**, enabling the prediction of real-time degradation behavior.

Overall, these kinetic models and equations provided a scientific basis for **calculating degradation rates, determining activation energy, identifying reaction order, and extrapolating stability data**, which were essential for predictive modeling in this study.

3.3 AI-Based Predictive Modeling:

AI-based predictive modeling was applied to estimate long-term drug degradation using short-term experimental data. The core principle used was the **linear (zero-order) degradation model**, where impurity increases proportionally with time:

Ct = m*months

- $\mathbf{C_t}$ = impurity at time *t*
- **m** = rate of impurity increase (slope)
- **t** = time (months)

This model was suitable because the experimental data showed a **linear increase in impurity over time**, indicating **zero-order kinetics**. Using this relationship, future impurity levels were extrapolated beyond the observed data.

Use of Excel Data

The experimental data (time, temperature, and impurity levels) were first organized in **Excel format**, which served as the input for the predictive model. This structured dataset enabled efficient processing, analysis, and model training.

Libraries and Their Purpose

To implement the predictive model, several Python libraries were used:

- **pandas(pd):**
  Used for **data handling and manipulation**. It allows easy reading of Excel files, organizing data into tables (dataframes), and preprocessing (such as cleaning column names). This was essential for structuring stability data before analysis.
- **numpy(np):**
  Used for **numerical operations and array handling**. It helped in defining time intervals (months) in a mathematical format suitable for model prediction.

- **scikit-learn(LinearRegression):**

  This is the core **machine learning component**. Linear Regression was used to:

  - Fit a straight line to impurity vs time data
  - Calculate the slope (**rate of degradation, m**)
  - Predict impurity values at future time points (extrapolation)

This directly aligns with the zero-order kinetic equation.

- **matplotlib.pyplot(plt):**

  Used for **data visualization**. It helped plot:

  - Actual experimental data points
  - Predicted degradation curves

    This visual comparison validates how well the model fits the data.

- **os:**

  Used for **file handling and automation**, such as:

  - Checking if the input file exists
  - Saving output files (Excel predictions and graphs) in the correct location

---

3.3.1 Python Implementation

Code 1: Data Import and Preprocessing (extrap.py)

This script imports and preprocesses experimental stability data from an Excel file. The dataset is read using the pandas library and verified before further analysis.

```
pip install pandas matplotlib
scikit-learn

import pandas as pd


# Load your file

data =
pd.read_excel("extrap.xlsx")


print(data.head()
```

This script ensures that the dataset is correctly loaded and structured. It provides a preview of the data and confirms the presence of required columns before applying predictive modeling.

Code 2: AI-Based Stability Prediction Model (stability_ai_tool.py)

This script implements an AI-based predictive model using linear regression to estimate impurity levels over time. The model is trained using experimental data and used to extrapolate future stability profiles under different temperature conditions.

```
import pandas as pd

from   sklearn.linear_model   import
LinearRegression

import matplotlib.pyplot as plt

import numpy as np

import os

print("\n📊 AI Stability Predictor")
```

(Complete code is provided in Appendix)

The script reads experimental data from an Excel file, validates the required parameters (time, temperature, impurity), and trains separate linear regression models for each temperature condition.

The model predicts impurity levels at future time points (up to 22 months) and generates both tabulated outputs and graphical representations. The results are automatically saved as an Excel file and a plot image, enabling efficient analysis and validation of degradation trends.

Working Principle of the Model

- The model reads experimental data from Excel
- Data is separated based on **temperature conditions**
- For each temperature:
    - A **linear regression model** is trained using time vs impurity

    - The model learns the degradation rate (**slope m**)
- Using this learned relationship, impurity values are **predicted for future months** (extrapolation)
- Results are:
    - Displayed as a table
    - Saved as an Excel file
    - Visualized as graphs

3.4 In Silico Prediction:

Zeneth is primarily known as a knowledge-based software application from Lhasa Limited that **predicts chemical degradation pathways** for small organic molecules, aiding pharmaceutical chemists with forced degradation studies. It helps identify **potential drug degradants and impurities** to assist in regulatory compliance and stability testing

Pathway Scores

Every predicted degradation product in Zeneth is assigned a Likelihood Score on a scale from 0 to 1000.

- **Scoring Tiers:**
    - 900 (Very Likely): High confidence that the degradation will occur.
    - 700 (Likely): Probable under the given conditions.
    - 500 (Equivocal): The degradation is possible but uncertain.
    - 300 (Unlikely): Low probability of occurrence.
    - 0 (Impossible): The reaction cannot happen under the selected conditions.

How the Score is Given

The score is calculated based on a combination of chemical rules and physicochemical descriptors:

- **Pattern Matching:** The initial score is derived from the "Transformation Dictionary," which contains hundreds of known degradation patterns. Each pattern has a base score reflecting how commonly that reaction is observed in scientific literature.
- **Condition Modulation:** The software adjusts the base score based on your inputs. For example, an ester hydrolysis might be assigned a score of 700 at pH 7, but this could jump to 900 if the pH is moved to an extreme (acidic or basic).
- **Physicochemical Predictors:** Zeneth uses internal tools to refine the score. For oxidation reactions, it may use a Bond Dissociation Energy (BDE) predictor to assess how easily a hydrogen atom can be abstracted. If the BDE is low, the likelihood score for that pathway increases.
- **Multi-generation Calculation:** For complex pathways where a degradant itself degrades further, the final pathway score is a product of the individual step scores, reflecting the decreased probability of long, multi-step sequences.

4. Results and discussions:

4.1 Extrapolated Stability Data:

The extrapolated stability data were generated to predict the long-term degradation behavior of the drug using short-term experimental data. Initially, extrapolation was performed from **6 months to 22 months** at **25°C and 40°C** using the zero-order degradation model:

Ct = m × months, where C_t represents impurity at time *t,* and **m** is the rate of impurity increase. Since the degradation followed a linear trend, this model allowed accurate prediction of impurity levels beyond the observed data

N1003		
	% of Impurity	
Time Point	Temp. 40°C	Temp. 25°C
0	0.05	0.05
3	0.7	0.2
6	1.4	0.3

N1003		
	% of Impurity	
Time Point	Temp. 40°C	Temp. 25°C
0	0.05	0.05
3	0.7	0.2
6	1.4	0.3
9	2.025	0.5
12	2.7	0.6
15	3.375	0.7
18	4.05	0.9
22	4.95	1.1

Each drug has a **predefined impurity limit** that must not be exceeded for safety and efficacy. By extending the degradation line, extrapolation helps identify the **time point (month)** at which this limit is likely to be crossed. This is critical for estimating **shelf life** and ensuring regulatory compliance.

Further, extrapolation was extended to **60°C for 6 months** using the Arrhenius-based relationship:

This equation enabled the prediction of degradation rate at higher temperatures using known values, supporting accelerated stability assessment.

ln{k_{60}}/{k{25}} = {E_a}/{R}({1}/{T{25}} - {1}/{T{60}}

where:

- **k_{60}, k_{25}** = rate constants at 60°C and 25°C
- **E_a** = activation energy
- **R** = gas constant
- **T** = temperature in Kelvin

N1003	
	% of impurity
Time Point	**Temp. 60°C**
0	**0.05**
3	**4.01**
6	**7.97**

4.2 Graphical Representation:

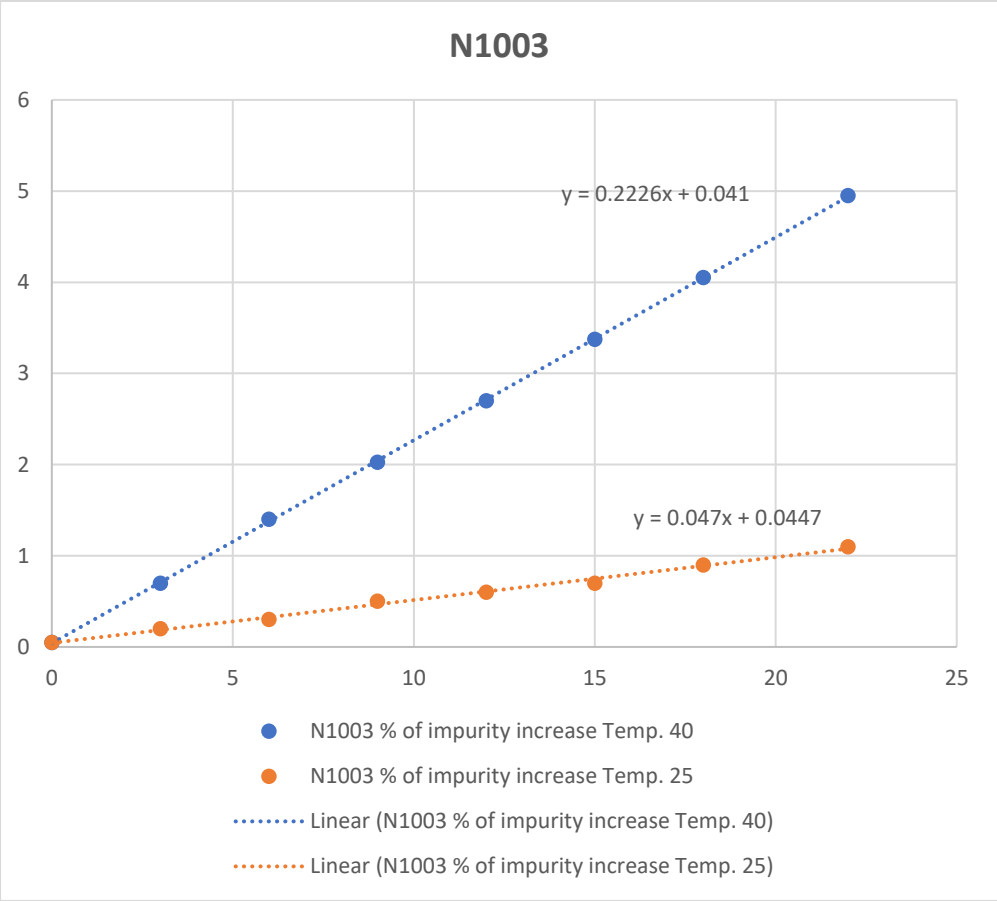
The extrapolated stability data were further represented graphically by plotting **impurity (%) versus time (months)** for different temperature conditions. The graphs showed a **linear increase in impurity over time**, confirming that the degradation follows **zero-order kinetics**.

From the plotted graphs, a straight line was obtained, and the **slope (m)** of this line represents the **rate of impurity increase**. This slope was directly used in the zero-order equation:

Ct= m × months

Thus, the graphical representation not only visually validates the degradation trend but also provides the **rate constant (m)** required for extrapolation.

Overall, the graph serves as both a **validation tool** for the kinetic model and a **quantitative source** for determining the degradation rate used in predictive calculations.



4.3 Zeneth Prediction Results:

The degradation behavior of the selected drug molecule was further analyzed using **Zeneth**. The molecular structure was entered into the software, and predictions of possible degradation pathways were generated at 25°C, 50°C, and 100°C over a **pH range of 1–14**.

To interpret the results, graphs were plotted for:

- Pathway score vs Temperature (at constant pH)
- Pathway score vs pH (at constant temperature)

The majority of the plots showed **horizontal linear trends**, indicating that the **pathway score remained consistently high (above 800)** across different temperatures and pH conditions. This suggests that a **single dominant degradation pathway** was followed, with no significant interference from alternative pathways. However, a **slight dip in pathway score was observed at pH 7** when the temperature was kept constant. This indicates minor sensitivity around neutral conditions and suggests involvement of **acid–base-catalyzed degradation behavior.**

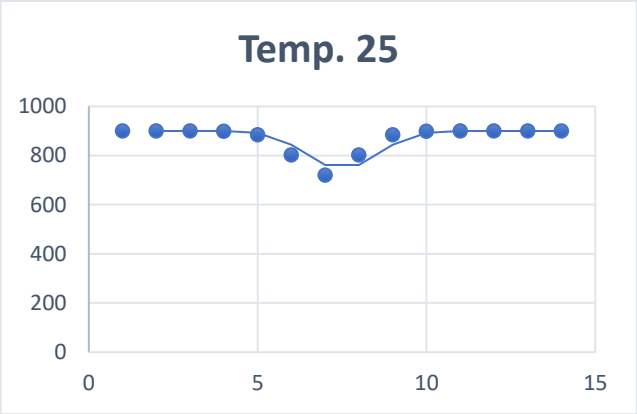
Overall, the consistently high and stable pathway scores confirm that degradation follows a uniform, **predictable pathway**. This strongly **validated the experimental findings**, supporting the conclusion that the observed impurity formation occurs through a specific and consistent degradation mechanism rather than multiple competing pathways.

EXACT MASS: 000.000

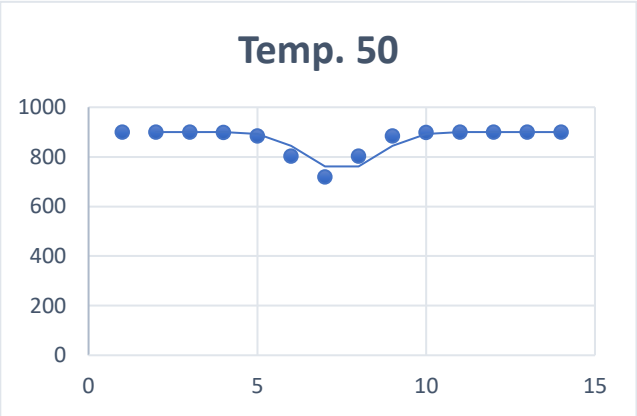
CHEMICAL FORMULA: ABC

TRANSFORMATION: HYDROLYSIS

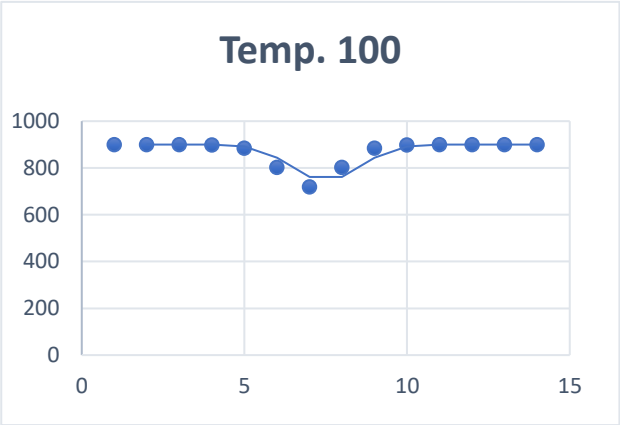
N1003	
Temp. 25°C	
X axis	Y axis
pH	Pathway score
1	900
2	900
3	900
4	899
5	885
6	803
7	720
8	803
9	885
10	899
11	900
12	900
13	900
14	900



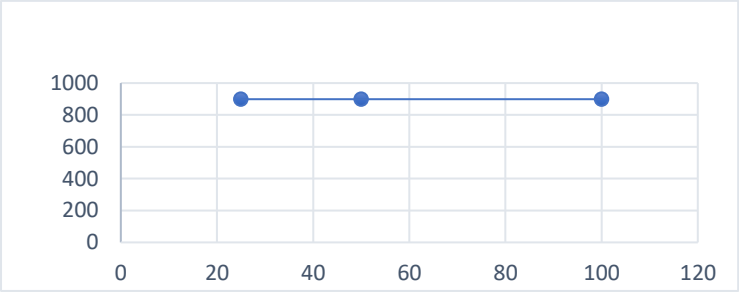
N1003	
Temp. 50°C	
X axis	Y axis
pH	Pathway score
1	900
2	900
3	900
4	899
5	885
6	803
7	720
8	803
9	885
10	899
11	900
12	900
13	900
14	900



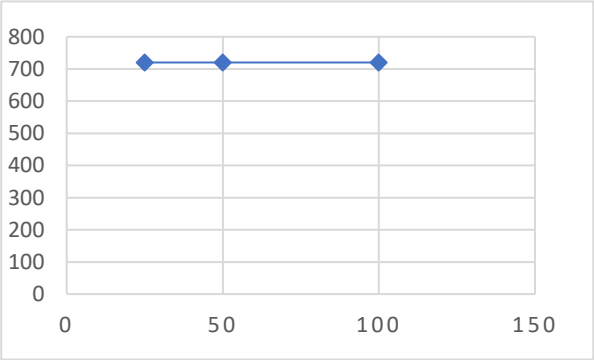
N1003	
Temp. 100°C	
X axis	Y axis
pH	Pathway score
1	900
2	900
3	900
4	899
5	885
6	803
7	720
8	803
9	885
10	899
11	900
12	900
13	900
14	900



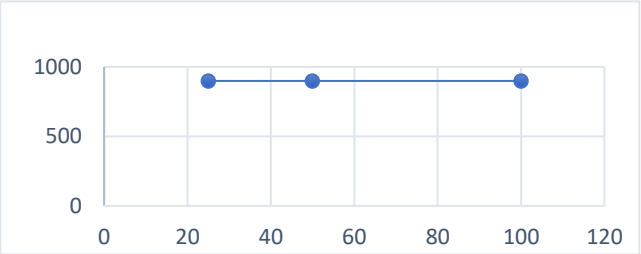
N1003	
pH 2	
X axis	Y axis
Temp(◦C)	Pathway score
25	900
50	900
100	900



N1003	
pH 7	
X axis	Y axis
Temp(◦C)	Pathway score
25	720
50	720
100	720



N1003	
pH 10	
X axis	Y axis
Temp(◦C)	Pathway score
25	899
50	899
100	899



4.4 AI Model Output:

The AI-based predictive model was used to estimate long-term impurity levels using the initial experimental data. The model was provided with **input data on impurity increase over the first 6 months under** different temperature conditions.

Using this input, the model applied **linear regression (zero-order kinetics) to learn the degradation pattern and predict future impurity levels**. As an output, the model generated **extrapolated impurity data up to 22 months** for each temperature condition.

In addition to tabulated results, the model also produced a **graph of impurity vs months**

The graph showed a **linear trend**, confirming that the model accurately captured the degradation behavior. The close alignment between actual and predicted values indicates that the AI model is reliable for **long-term stability prediction**.

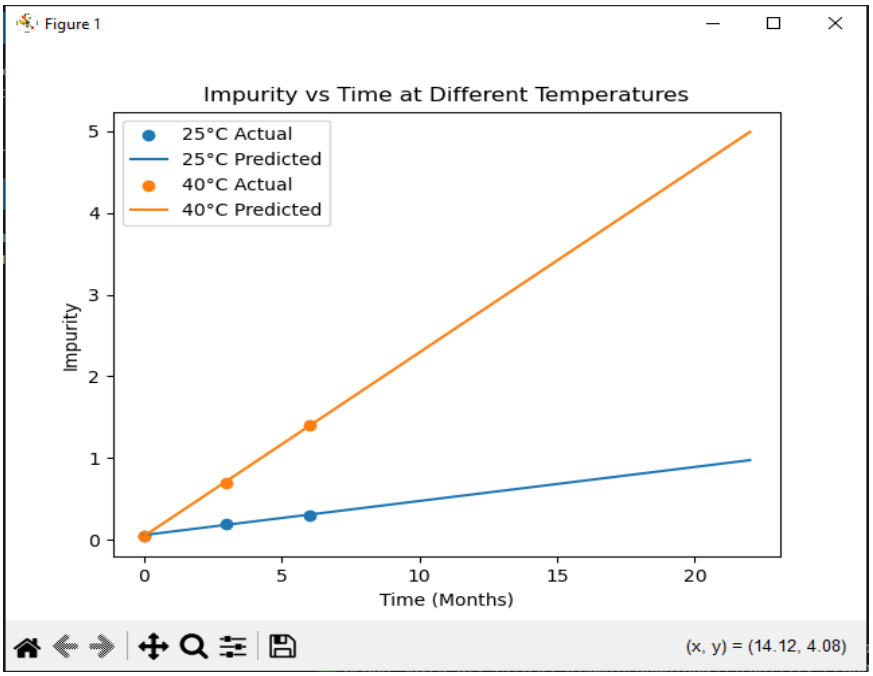
Overall, this approach demonstrates how AI can effectively extend short-term stability data to predict future degradation, thereby supporting faster, more efficient pharmaceutical analysis.

INPUT

N1003		
	% of Impurity	
Time Point	Temp. 40°C	Temp. 25°C
0	0.05	0.05
3	0.7	0.2
6	1.4	0.3

OUTPUT

```
Prediction Table:
   Months  Impurity_25C  Impurity_40C
0       0      0.058333      0.041667
1       3      0.183333      0.716667
2       6      0.308333      1.391667
3       9      0.433333      2.066667
4      12      0.558333      2.741667
5      15      0.683333      3.416667
6      18      0.808333      4.091667
7      22      0.975000      4.991667
```



4.5 Comparative Analysis:

The comparative analysis was conducted in two parts to assess the consistency and reliability of the predictive approaches used in this study.

4.5.1 Comparison of Extrapolation Methods (Experimental vs Excel vs AI)

A comparison was made between extrapolated data obtained from:

- Data
- Excel-based extrapolation
- AI-based predictive modeling

The results from all three approaches showed **close agreement**, with negligible variation in predicted impurity values over the 22-month period. This indicates that:

- The degradation followed a **consistent linear (zero-order) trend**
- The **mathematical model (Ct = m × t)** was correctly applied
- The **AI model successfully learned and replicated the degradation pattern**

This strong correlation validates the **accuracy and reliability of the AI-based model**, confirming that it can effectively be used for long-term stability prediction.

% OF IMPURITY AT pH3.5						
	DATA		EXCEL		AI	
Time point (in months)	Temp. 40°C	Temp. 25°C	Temp. 40°C	Temp. 25°C	Temp. 40°C	Temp. 25°C
0	0.015	0.015	0.015	0.015	0.015	0.015
1	0.1	ND	0.1	0.0207	0.1	0.0217
2	0.191	ND	0.191	0.0414	0.191	0.0419
3	0.281	0.077	0.281	0.077	0.2801	0.07034
6	0.587	0.152	0.5334	0.1242	0.5468	0.1949

% OF IMPURITY AT pH 5						
	DATA		EXCEL		AI	
Time point (in months)	Temp. 40°C	Temp. 25°C	Temp. 40°C	Temp. 25°C	Temp. 40°C	Temp. 25°C
0	0.057	0.057	0.057	0.057	0.057	0.057
1	0.489	ND	0.489	0.0915	0.489	0.0957
2	0.885	ND	0.885	0.183	0.885	0.203
3	1.296	0.319	1.296	0.2745	1.296	0.319
6	2.449	0.643	2.4018	0.549	2.5326	0.5575

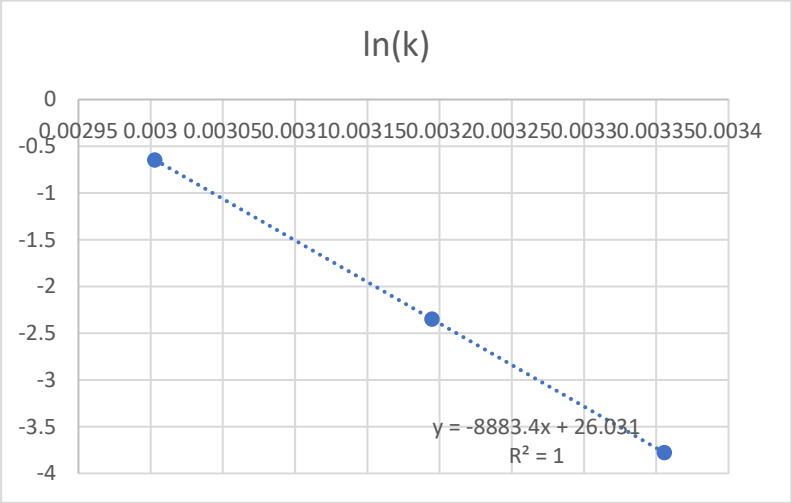
4.5.2 Comparison of Activation Energy Calculations and Zeneth

Activation energy (**E_a**) was calculated using Arrhenius-based analysis for different pH conditions. The values obtained were nearly identical **across all pH levels**. They were calculated from extrapolated concentration values.

Using the formulas:

- **k = {C_0 - C_t}/{t}**
- **ln{k_{60}}/{k{25}}={E_a}/{R}({1}/{T{25}}-{1}/{T{60}}**

AT pH 3.5				
TEMP(◦C)	**TEMP(K)**	**K**	**1/T(K^{-1})**	**ln k**
25	**298**	**0.02283**	**0.003356**	**-3.779**
40	**313**	**0.09533**	**0.003195**	**-2.3504**
60	**333**	**0.5242**	**0.003003**	**-0.6458**



Ea = m*8.314

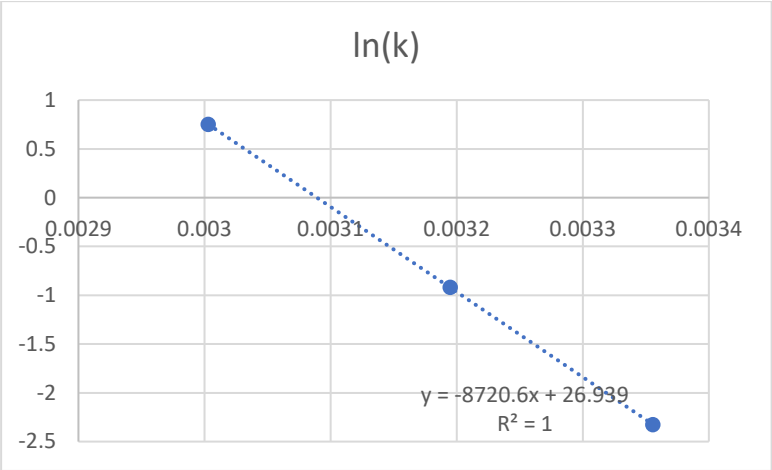
where, m – slope

8.314 – gas constant

Ea = -(-8883.4*8.314)

Ea = 73.85 kJ/mol

AT pH 5				
TEMP(◦C)	**TEMP(K)**	**K**	**1/T(K^{-1})**	**ln k**
25	**298**	**0.097**	**0.0033557**	**-2.326**
40	**313**	**0.398**	**0.0031949**	**-0.9196**
60	**333**	**2.117000017**	**0.003003**	**0.75**



Ea = m*8.314

Ea = -(-8883.4*8.314)

Ea = 72.70 kJ/mol

- Since **activation energy remains almost constant**, it indicates that:
  - The reaction requires a **similar amount of energy** for degradation
  - The **rate-determining step remains unchanged**
  - The **degradation mechanism is consistent across pH conditions**

  Reason for this observation:

  The degradation is likely governed by a **single dominant pathway** that is not significantly influenced by pH changes. While pH may slightly affect the rate (minor variations), it does not alter the **reaction's energy barrier**

(E_a). This suggests that the **same transition state and mechanism** are involved in impurity formation under different pH conditions.

Correlation with Zeneth Prediction

The findings were further supported by **Zeneth's** predictions, which showed:

- **Consistently high pathway scores (>800)**
- **Minimal variation across temperature and pH**
- A **single dominant degradation pathway**

This confirms that:

- No alternative degradation mechanisms significantly contributed
- The reaction followed a **uniform and stable pathway**

Overall Conclusion from Comparative Analysis

The agreement between experimental data, Excel-based calculations, AI predictions, Arrhenius analysis, and **Zeneth** results clearly demonstrates that:

- The degradation follows a **consistent zero-order kinetic model**
- The **activation energy is independent of pH**, indicating a stable degradation mechanism
- A **single dominant degradation pathway** governs impurity formation

Most importantly, this study successfully establishes a strong connection between **in silico prediction and real-time experimental data**. The degradation pathway predicted using Zeneth closely matched the experimentally observed impurity formation, while the AI-based model accurately reproduced and extrapolated real stability data.

This confirms that **in silico tools can reliably predict degradation pathways and behavior**, reducing dependence on long-term experimental studies. The close correlation between predicted and actual data validates the

effectiveness of combining **Arrhenius kinetics, AI modeling, and computational tools** for stability assessment.

Thus, the overall work concludes that **in silico prediction of degradation pathways, when supported by experimental validation, can serve as a powerful and efficient approach for pharmaceutical stability studies, where Zeneth effectively acted as a real-time comparative tool to validate and confirm the experimental degradation pathway.**

5. Conclusion:

This study focused on the **in silico prediction of pharmaceutical degradation pathways** and their comparison with experimental data to evaluate the reliability of predictive approaches in stability studies.

Through this work, degradation data obtained for the first 6 months were successfully **extrapolated up to 22 months** using both **Excel-based calculations and AI-based modeling**. The close agreement between these methods and the experimental data confirmed the accuracy of the extrapolation approach and validated the use of zero-order kinetics to describe the degradation behavior.

Further, **activation energy (E_a)** was calculated using Arrhenius analysis across different pH conditions. The similarity in activation energy values indicated that the **degradation process follows a consistent mechanism**, with no significant change in the energy barrier across pH values.

In addition, **Zeneth** predictions demonstrated a **single dominant degradation pathway** with consistently high pathway scores. This aligned closely with the experimental observations, confirming that the degradation proceeds through a **uniform and stable pathway**, free of interference from alternative mechanisms.

Overall, this study concludes that **in silico prediction using extrapolated data and activation energy calculations can effectively model pharmaceutical degradation behavior**. The strong correlation between experimental data, AI predictions, and Zeneth results highlights that **computational tools can reliably predict and validate degradation pathways**, with Zeneth serving as a **real-time comparative platform to confirm the experimentally observed pathway**.

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5. Appendix: stability_ai_tool.py

```
import pandas as pd
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
import numpy as np
import os
print("\n📊 AI Stability Predictor")
print("=====================================")
# 🔷 Ask for file path
file_path = input("📂 Enter Excel file path: ")
# 🔷 Check if file exists
if not os.path.exists(file_path):
    print("❌ File not found. Check path again.")
    exit()
# 🔷 Load data
df = pd.read_excel(file_path)
df.columns = df.columns.str.lower()
print("\nData Preview:")
print(df.head())

# 🔷 Check required columns
required = ['time', 'temperature', 'impurity']
if not all(col in df.columns for col in required):
    print("❌ Your Excel must have columns: TIME, TEMPERATURE, IMPURITY")
    exit()
temps = df['temperature'].unique()
models = {}
results = {}
months = np.array([0, 3, 6, 9, 12, 15, 18, 22]).reshape(-1, 1)
plt.figure()
```

```
# 🔷 Train + predict
for temp in temps:
    temp_df = df[df['temperature'] == temp]
    model = LinearRegression()
    model.fit(temp_df[['time']], temp_df['impurity'])
    predictions = model.predict(months)
    models[temp] = model
    results[temp] = predictions
    # Plot
    plt.scatter(temp_df['time'], temp_df['impurity'], label=f"{temp}°C
Actual")
    plt.plot(months, predictions, label=f"{temp}°C Predicted")


# 🔷 Create table
table = pd.DataFrame({'Months': months.flatten()})
for temp in temps:
    table[f'Impurity_{temp}C'] = results[temp]
print("\n📋 Prediction Table:")
print(table)
# 🔷 Save outputs automatically
output_folder = os.path.dirname(file_path)
table_path = os.path.join(output_folder, "prediction_output.xlsx")
graph_path = os.path.join(output_folder, "stability_graph.png")
table.to_excel(table_path, index=False)
plt.savefig(graph_path)
print(f"\n✅ Table saved at: {table_path}")
print(f"✅ Graph saved at: {graph_path}")
# 🔷 Show graph
plt.xlabel("Time (Months)")
plt.ylabel("Impurity")
plt.title("Impurity vs Time at Different Temperatures")
plt.legend()
plt.show()
```